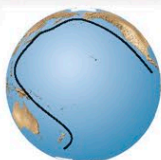


**FIERY ARENAL**

Arenal is one of the most active volcanoes in Costa Rica—a region where the small Cocos Plate is subducted under the neighboring Caribbean Plate.

PACIFIC OCEAN**Pacific Ring of Fire**

LOCATION Pacific Ocean rim, from Chile to New Zealand

TYPE Series of destructive boundaries

LENGTH 20,000 miles (32,000 km)

The Ring of Fire is a huge arc of volcanic and seismic (earthquake) activity around the rim of the Pacific Ocean. It stretches from the western coasts of South America and North America, across the Aleutian Islands of Alaska, and down the eastern edge of Asia, to the northeast of Papua New Guinea, and finally to New Zealand. More than half of the world's active volcanoes above sea level are part of the ring. The Ring of Fire results from the Pacific Plate

and other smaller plates in the Pacific colliding with neighboring plates along a series of destructive plate boundaries. The main driving force for this activity is the creation of new crust by a large midocean ridge in the eastern Pacific (the East Pacific Rise). Here, new material is continually added to the Pacific and Nazca Plates and to the small Cocos Plate, forcing them toward the edges of the Pacific.

Across much of its northern and western edges, the oceanic crust of the Pacific Plate is subducted (forced underneath) by the oceanic crust of other plates, forming deep-sea trenches. This predisposes these regions to earthquakes, and the subducted crust also melts deep down to create hot magma, which reaches the surface through volcanoes. The result has been the formation of many highly volcanic island arcs in these regions—examples being the Aleutian Islands, the Kurile Islands, the islands of Japan, and the Mariana Islands.

On the eastern side of the Pacific, the situation is somewhat different.

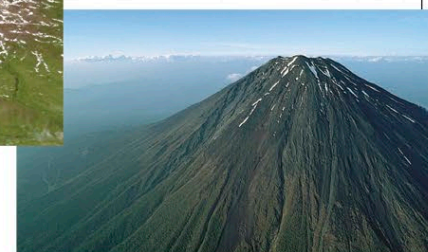
Here, parts of the Pacific, Nazca, and Cocos Plates are being subducted below continental crust. Deep-sea trenches have also formed here, but instead of island arcs, the plate collisions

THE ANDES

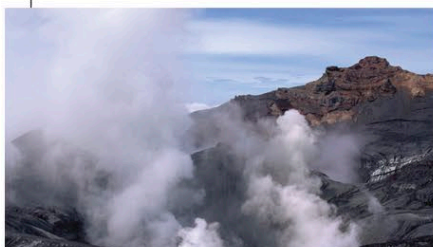
On the western edge of South America, subduction of the Nazca Plate under the South American Plate has created the Andes, another highly active region.

**OKMOK VOLCANO**

The volcanic Aleutian Islands were created as the Pacific Plate was driven under the oceanic crust of the North American Plate. This volcano is on the island of Umnak.

**MOUNT FUJI**

In the northwest Pacific, the subduction of the Pacific Plate under the Eurasian Plate is responsible for creating the islands of Japan, the site of volcanoes such as Mount Fuji, which last erupted in 1707.

**MOUNT RUAPEHU**

At the southwest corner of the Ring of Fire is New Zealand. Here, steam rises from the country's tallest volcano, Ruapehu, between eruptions that occurred in 1995 and 1996.